

Math 55 quiz 03/03/2026

You must type or neatly handwrite each answer on a separate piece of paper and select the pages for each question when you upload to Gradescope. Feel free to use more than one sheet for an answer if needed. Note that both PDF and picture/photo format (JPG) are accepted. You must show your work. During the quiz, you may consult the textbook, your own notes and other course materials you yourself prepared before the beginning of the quiz. You may not consult resources online. You can post a private question to Ed Discussion to instructors only, but do not make public posts on Ed Discussion, Discord, or other online forums.

By submitting this quiz, you agree to adhere to the statement below: “As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. All of the work submitted here is mine, and I have not discussed the content of this exam with any other person. I have not consulted any resources beyond the course materials and my notes during the exam. I have not used calculators, computers, or the internet to do calculations or look up answers.”

1. Prove using strong induction that every positive integer can be written as a sum of distinct powers of 2. Explain where this proof fails if you replace 2 with 3. (Note that it's obvious that 2 cannot be written as a sum of distinct powers of 3; you're being asked to explain why the *proof* wouldn't work, not just show that the *claim* is false.)

Solution: Let n be the integer. We use strong induction. Base case: $n = 1$ is just 2^0 . Now, if n is a power of 2, we're done. Otherwise, take the largest power of 2 less than n . Call it k . Then $n - 2^k$ can be written as a sum of distinct powers of 2, by strong induction, and none of those powers are k or larger, since otherwise we would have $n \geq 2^k \times 2 > n$. This is where the proof fails for powers of 3: $n - 3^k$ isn't necessarily less than 3^k .

2. For any positive integer n , define $a \Downarrow n$ recursively as follows:

- $a \Downarrow 1 = a$
- $a \Downarrow n = (a \Downarrow (n - 1))^a$

Prove that for all $n \geq 1$, $a \Downarrow n = a^{a^{n-1}}$.

Solution: Our base case is straightforward. For the inductive step, assume that $a \Downarrow k = a^{a^{k-1}}$. Then by definition, $a \Downarrow (k + 1) = (a \Downarrow k)^a = a^{a^k}$.